New display that lets users touch and feel objects on screen

A team led by Prof. M. Manivannan, Department of Applied Mechanics, IIT Madras has developed a new touchscreen display that allows users to feel the textures of objects on the display, called iTad (interactive touch active display). It can mimic textures like crisp edges and gritty surfaces. The display does not consist of any moving parts; it integrates multi-touch sensing with haptics on the same layer. It comprises an in-built multi-touch sensor that detects finger movement and the surface friction is adjusted via software. The electric field is controlled using a physical phenomenon known as electro adhesion; the software changes friction locally as fingers travel across a smooth plane.

The technology iTad uses electroadhesion to mimic textures such as crisp edges and gritty surfaces (Source: https://www.iitm.ac.in)

A tool that could help understand brain health and disease

Researchers at Brigham have developed a powerful computational tool for understanding brain health and disease that provides an enhanced way of fingerprinting sleeping brain activity. The new method extracts tens of thousands of electrical events from the brain waves of a sleeping person. Information from these waveforms is then used to create a picture of brain activity that is like a fingerprint, which is unique for each person and consistent from one night to the next. Instead of looking at the waveforms in terms of fixed sleep stages as standard sleep studies do, the researchers characterised the full continuum of gradual changes that occur in the brain during sleep. This study could be useful in better understanding of brain health and disease.

Unique brain wave 'fingerprints' yield pictures of sleeping brain to target disease (Source: https://hms.harvard.edu)